A 1.5 V, 50 mA PMOS Low-Dropout Regulator in 0.18 μm CMOS

Part 1: Theory and proofs of the design formulas

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Abstract

This note derives, with proofs, every formula used to design the low-dropout (LDO) regulator of Fig. 1. It covers the device equations (square law, triode region, weak inversion), the loop polarity, the static output equation (which gives accuracy, load and line regulation in one line), the pass-device and headroom conditions, the loop gain with its poles and zero, the unity-gain frequency and phase margin, the ESR window, the power-supply rejection and the load-step response. Part 2 (`ldo_hand_design.pdf`) applies these formulas to a specification to size every transistor and checks the result in ngspice.

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1 Architecture and notation

Architecture (Fig. 1).

- *Pass device* M_P: a PMOS. A PMOS can work with $V_{IN} - V_{OUT}$ down to its saturation voltage, which is what makes a low dropout possible.
- *Feedback divider* R_1, R_2: it feeds back the fraction $\beta = R_2/(R_1 + R_2)$ of V_{OUT}.
- *Error amplifier* (EA) M_1–M_5: a five-transistor OTA with a PMOS input pair, which works at $V_{REF} = 0.6\,\text{V}$. An NMOS pair would need its input common mode above $V_{GS} + V_{DS,sat} \approx 0.75\,\text{V}$.
- *Buffer* M_B: a PMOS source follower with its bulk tied to its source, biased by M_{BS}. It drives the large gate capacitance of M_P from a low impedance (Section 4.2).
- *Load*: the off-chip capacitor C_L with its series resistance R_{ESR}. The load current is an ideal current sink.

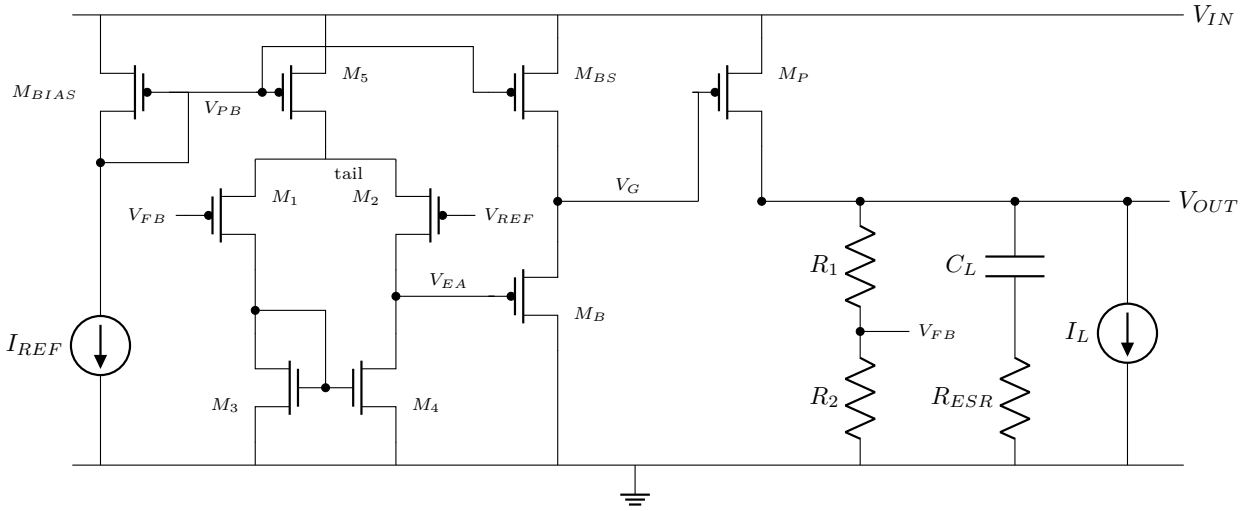


Figure 1: LDO schematic. M_1 (gate V_{FB}) sits on the diode side of the mirror, M_2 (gate V_{REF}) on the output side; Lemma 4 shows that this gives negative feedback. Bulk of every PMOS is V_{IN} except M_1, M_2 (tail node) and M_B (its source).

2 Device equations

PMOS voltages are written as positive magnitudes (V_{SG}, V_{SD}, $|V_{TH}|$). The overdrive is $V_{OV} = V_{GS} - V_{TH}$ (NMOS) or $V_{SG} - |V_{TH}|$ (PMOS), and $K' = \mu C_{ox}$.

Lemma 1 (Square law and its small-signal parameters)**.** *In saturation ($V_{DS} \geq V_{OV} > 0$),*

$$I_D = \tfrac{1}{2} K' \tfrac{W}{L} V_{OV}^2 \,(1 + \lambda V_{DS}). \tag{1}$$

Neglecting λV_{DS} against 1 in the result,

$$g_m = \sqrt{2K' \tfrac{W}{L} I_D} = \frac{2I_D}{V_{OV}}, \qquad g_{ds} = \lambda I_D, \qquad r_o = \frac{1}{\lambda I_D}. \tag{2}$$

Proof. *Step 1.* $g_m = \partial I_D / \partial V_{GS} = K'(W/L)\, V_{OV}$, by differentiating (1) with $\lambda V_{DS} \ll 1$.

Step 2. From (1), $V_{OV} = \sqrt{2I_D/(K'W/L)}$. Substituting into Step 1 gives $g_m = \sqrt{2K'(W/L) I_D}$.

Step 3. Dividing $g_m = K'(W/L)V_{OV}$ by $I_D = \tfrac{1}{2}K'(W/L)V_{OV}^2$ gives $g_m/I_D = 2/V_{OV}$.

Step 4. $g_{ds} = \partial I_D/\partial V_{DS} = \tfrac{1}{2}K'\tfrac{W}{L}V_{OV}^2\lambda = \lambda I_D/(1+\lambda V_{DS}) \approx \lambda I_D$. □

Lemma 2 (Triode region)**.** *For $0 \leq V_{DS} \leq V_{OV}$, $I_D = K'\tfrac{W}{L}\big(V_{OV}V_{DS} - \tfrac{1}{2}V_{DS}^2\big)$, so $g_m = K'\tfrac{W}{L}V_{DS}$ and $g_{ds} = K'\tfrac{W}{L}(V_{OV} - V_{DS})$. In particular*

$$\frac{g_{ds}}{g_m} = \frac{V_{OV} - V_{DS}}{V_{DS}}, \tag{3}$$

which grows without bound as V_{DS} falls below V_{OV}.

Proof. Differentiate I_D with respect to V_{GS} (only V_{OV} depends on it) and with respect to V_{DS}, then divide the two results. □

Lemma 3 (Weak inversion)**.** *For V_{GS} below V_{TH}, $I_D = I_0 \exp\big(V_{GS}/(nV_T)\big)$ with slope factor $n \approx 1.3$–1.5 and $V_T = kT/q$. Then $g_m = I_D/(nV_T)$.*

Proof. $g_m = \partial I_D/\partial V_{GS} = I_0 e^{V_{GS}/(nV_T)}/(nV_T) = I_D/(nV_T)$. □

3 Large-signal (DC) analysis

3.1 Loop polarity

Lemma 4 (Negative feedback)**.** *With V_{FB} on M_1 (diode side of the mirror) and V_{REF} on M_2, a rise of V_{OUT} lowers the pass-device current.*

Proof. *Step 1.* If V_{OUT} rises, $V_{FB} = \beta V_{OUT}$ rises.

Step 2. Then V_{SG1} falls, so I_{D1} falls, while $I_{D2} = I_{tail} - I_{D1}$ rises.

Step 3. The mirror makes M_4 sink a copy of I_{D1}. The output node therefore receives I_{D2} and loses only $I_{D1} < I_{D2}$, so V_{EA} rises.

Step 4. The source follower gives $V_G = V_{EA} + V_{SG,B}$, so V_G rises.

Step 5. $V_{SG,P} = V_{IN} - V_G$ falls, so the current of M_P falls, which opposes the rise of V_{OUT}. □

3.2 The static output equation

Linearise the EA around the balanced point. When $V_{FB} = V_{REF}$, the two halves carry equal currents, so $V_{DS4} = V_{DS3} = V_{GS3}$ and $V_{EA} = V_{GS3}$. Writing A_{EA} for the EA gain and using Lemma 4 for the sign gives

$$V_{EA} = V_{GS3} + A_{EA}\,(V_{FB} - V_{REF}). \tag{4}$$

Proposition 5 (Static output equation)**.** *Let $V_{SG,P}(I)$ be the source–gate voltage that makes M_P carry $I = I_L + I_{div}$. Then*

$$\boxed{V_{OUT} = \frac{V_{REF}}{\beta} + \frac{\varepsilon}{\beta A_{EA}}, \qquad \varepsilon \equiv V_{IN} - V_{SG,P}(I_L + I_{div}) - V_{SG,B} - V_{GS3}.} \tag{5}$$

Proof. *Step 1.* The gate of M_P sits at $V_G = V_{IN} - V_{SG,P}$, because M_P must carry $I_L + I_{div}$.

Step 2. The follower gives $V_{EA} = V_G - V_{SG,B} = V_{IN} - V_{SG,P} - V_{SG,B}$.

Step 3. Set this equal to (4) with $V_{FB} = \beta V_{OUT}$: $V_{IN} - V_{SG,P} - V_{SG,B} = V_{GS3} + A_{EA}(\beta V_{OUT} - V_{REF})$.

Step 4. Solve for V_{OUT}: $\beta A_{EA} V_{OUT} = A_{EA} V_{REF} + (V_{IN} - V_{SG,P} - V_{SG,B} - V_{GS3})$.

Step 5. Divide by βA_{EA}. □

Proposition 5 contains three specifications at once.

Corollary 6. *(a)* Ideal value: *as $A_{EA} \to \infty$, $V_{OUT} \to V_{REF}/\beta = V_{REF}(1 + R_1/R_2)$.*

(b) Load regulation: *only $V_{SG,P}$ depends on I_L, so $\Delta V_{OUT} = -\Delta V_{SG,P}/(\beta A_{EA})$ between two loads.*

(c) Line regulation: *with $V_{SG,P}$, $V_{SG,B}$, V_{GS3} fixed by their currents, $\partial V_{OUT}/\partial V_{IN} = 1/(\beta A_{EA})$.*

(d) Systematic offset: *ε is the difference $V_{DS4} - V_{DS3}$ that the loop has to create. The output error at a given load is $\varepsilon/(\beta A_{EA})$, and it changes sign between light and heavy load.*

When the follower gain $A_B < 1$ is kept (Section 4.2), A_{EA} becomes $A_{EA}A_B$ in (b) and (c).

3.3 Pass device: saturation and dropout

Proposition 7 (Pass-device sizing)**.** *(a) M_P stays saturated at $I_L = I_{max}$ and $V_{IN} = V_{IN,nom}$ if*

$$\frac{W}{L} \geq \frac{2I_{max}}{K'\,(V_{IN,nom} - V_{OUT})^2}. \tag{6}$$

(b) At $V_{SD} = V_{DO}$ (triode), carrying I_{max} needs the overdrive

$$V_{OV} = \frac{I_{max}}{K'\frac{W}{L}V_{DO}} + \frac{V_{DO}}{2}. \tag{7}$$

The dropout specification is met if the drive chain can pull V_G down to $V_{IN} - |V_{TH}| - V_{OV}$.

Proof. *Step 1.* (a) Saturation needs $V_{SD} \geq V_{OV}$, and $V_{SD} = V_{IN,nom} - V_{OUT}$.

Step 2. Use (1) at the edge $V_{OV} = V_{SD}$: $I_{max} \leq \frac{1}{2}K'\frac{W}{L}(V_{IN,nom} - V_{OUT})^2$. Rearranging gives (6).

Step 3. (b) Lemma 2 at $V_{DS} = V_{DO}$: $I_{max} = K'\frac{W}{L}(V_{OV}V_{DO} - \frac{1}{2}V_{DO}^2)$.

Step 4. Divide by $K'\frac{W}{L}V_{DO}$ and move $V_{DO}/2$ to the right-hand side. This is (7). □

3.4 Headroom window of the error-amplifier output

Proposition 8 (Buffer V_{SG} window)**.** *M_2 and M_4 stay saturated over the whole input and load range if*

$$\underbrace{V_{IN,max} - V_{SG,P}(I_{min}) - V_{REF} - |V_{TH,p}|}_{lower\ bound} \;\le\; V_{SG,B} \;\le\; \underbrace{V_{IN,min} - V_{SG,P}^{DO}(I_{max}) - V_{OV4}}_{upper\ bound}, \tag{8}$$

where $V_{SG,P}^{DO}$ is the pass-device V_{SG} at the dropout point.

Proof. *Step 1.* M_2 saturated: $V_{SD2} \ge V_{OV2}$, i.e. $V_{tail} - V_{EA} \ge V_{SG2} - |V_{TH,p}|$.

Step 2. The tail sits one V_{SG} above the gate of M_2: $V_{tail} = V_{REF} + V_{SG2}$. Substitute: $V_{EA} \le V_{REF} + |V_{TH,p}|$.

Step 3. M_4 saturated: $V_{DS4} = V_{EA} \ge V_{OV4}$.

Step 4. From the proof of Proposition 5, $V_{EA} = V_{IN} - V_{SG,P} - V_{SG,B}$.

Step 5. V_{EA} is largest at $V_{IN,max}$ and light load (small $V_{SG,P}$). Insert into Step 2 and solve for $V_{SG,B}$: this is the lower bound.

Step 6. V_{EA} is smallest at $V_{IN,min}$ and full load. Insert into Step 3 and solve for $V_{SG,B}$: this is the upper bound. □

4 Small-signal loop analysis

4.1 Loop gain

Proposition 9 (Loop gain)**.** *Break the loop at the gate of M_1 and inject v_t there. Then*

$$T(s) \equiv -\frac{v_{fb}}{v_t} = \beta\, A_{EA}(s)\, A_B(s)\, g_{m,P}\, Z_o(s), \qquad Z_o(s) = R_o \,\Big\|\, \Big(R_{ESR} + \frac{1}{sC_L}\Big), \quad R_o = r_{o,P} \,\|\, (R_1 + R_2). \tag{9}$$

Proof. *Step 1.* The gate of M_1 draws no current, so breaking the loop there leaves the divider unloaded and the break is exact.

Step 2. EA: $v_{ea} = A_{EA}(s) v_t$ (Lemma 10 below; the sign is positive by Lemma 4).

Step 3. Buffer: $v_g = A_B(s) v_{ea}$ (Lemma 11).

Step 4. Pass device, with $v_{in} = 0$: the drain current into the output node is $i = g_{m,P} v_{sg} = -g_{m,P} v_g$.

Step 5. Output node: $v_{out} = i\, Z_o$. Here Z_o is the drain resistance, the divider and the capacitor branch in parallel; the ideal current-sink load adds nothing.

Step 6. Divider: $v_{fb} = \beta v_{out} = -\beta g_{m,P} Z_o A_B A_{EA} v_t$.

Step 7. Divide by $-v_t$. □

Lemma 10 (Five-transistor OTA)**.** *At low frequency, $A_{EA} = \dfrac{g_{m1}}{g_{ds2} + g_{ds4}}$, and the output pole is $p_{EA} = \dfrac{1}{2\pi R_{o,EA} C_{EA}}$ with $R_{o,EA} = 1/(g_{ds2} + g_{ds4})$, where C_{EA} is the total capacitance at the EA output.*

Proof. *Step 1.* Ideal tail: $i_1 + i_2 = 0$, where i_k is the small-signal drain current of M_k toward the mirror.

Step 2. $i_1 = -g_m(v_{fb} - v_x)$ and $i_2 = -g_m(v_{ref} - v_x)$, with v_x the tail voltage (PMOS: a higher gate means less current).

Step 3. Adding the two and using Step 1 gives $v_x = (v_{fb} + v_{ref})/2$, so $i_1 = -g_m v_d/2$ with $v_d = v_{fb} - v_{ref}$.

Step 4. The mirror sinks i_1 at the output, and M_2 sources $i_2 = -i_1$. The net current into the output node is $i_2 - i_1 = -2i_1 = g_m v_d$.

Step 5. This current flows into $g_{ds2} + g_{ds4}$ in parallel with sC_{EA}: $v_{ea} = g_m v_d/(g_{ds2} + g_{ds4} + sC_{EA})$.

Step 6. At $s = 0$ this is A_{EA}. The single pole sits at $(g_{ds2} + g_{ds4})/(2\pi C_{EA})$. The mirror pole g_{m3}/C_{d1} is far higher and is neglected. □

Lemma 11 (PMOS source follower)**.** *With the bulk tied to the source and C_G the load capacitance at V_G,*

$$A_B(s) = \frac{g_{mB}}{g_{out} + sC_G}, \qquad g_{out} = g_{mB} + g_{dsB} + g_{ds,BS}, \qquad A_B = \frac{g_{mB}}{g_{out}}, \quad p_B = \frac{g_{out}}{2\pi C_G}. \tag{10}$$

Proof. *Step 1.* M_B carries $i = g_{mB}v_{sg} + g_{dsB}v_{sd} = g_{mB}(v_g - v_{ea}) + g_{dsB}v_g$ from V_G to ground. No g_{mb} term appears because $v_{bs} = 0$.

Step 2. M_{BS} (gate fixed) delivers $-g_{ds,BS}v_g$ into the node.

Step 3. KCL at V_G: $-g_{ds,BS}v_g = g_{mB}(v_g - v_{ea}) + g_{dsB}v_g + sC_Gv_g$.

Step 4. Collect the v_g terms: $v_g(g_{out} + sC_G) = g_{mB}v_{ea}$. □

Remark 12 (Which capacitance loads V_G). $C_G = C_{gs,P} + C_{gd,P}$, with no Miller multiplication of $C_{gd,P}$. The reason is that at the frequencies where p_B matters (near f_u), C_L already holds the drain of M_P at AC ground, so the gain from gate to drain is about zero. In saturation, $C_{gs} \approx \frac{2}{3}WLC_{ox} + WC_{ov}$ and $C_{gd} \approx WC_{ov}$.

4.2 Poles, zero and unity-gain frequency

Lemma 13 (Output pole and ESR zero)**.**

$$Z_o(s) = R_o\,\frac{1 + sR_{ESR}C_L}{1 + s(R_o + R_{ESR})C_L}, \qquad p_o = \frac{1}{2\pi(R_o + R_{ESR})C_L}, \qquad z_{ESR} = \frac{1}{2\pi R_{ESR}C_L}. \tag{11}$$

Proof. $Z_o = R_o(R_{ESR} + 1/sC_L)/(R_o + R_{ESR} + 1/sC_L)$. Multiply the numerator and the denominator by sC_L. □

Proposition 14 (Unity-gain frequency)**.** *If $p_o \ll f_u \ll z_{ESR}, p_B, p_{EA}$, then*

$$\boxed{f_u \approx \frac{G\,g_{m,P}}{2\pi C_L}, \qquad G \equiv \beta A_{EA}A_B,} \tag{12}$$

independent of R_o. Since $g_{m,P}$ rises with I_L, the highest f_u occurs at full load.

Proof. *Step 1.* For $f \gg p_o$, $|sC_L| \gg 1/R_o$; for $f \ll z_{ESR}$, $R_{ESR} \ll 1/|sC_L|$. Hence $Z_o \approx 1/(sC_L)$.

Step 2. For $f \ll p_B, p_{EA}$, $A_{EA}(s)A_B(s) \approx A_{EA}A_B$.

Step 3. By (9), $|T(j2\pi f)| \approx Gg_{m,P}/(2\pi fC_L)$. Setting this equal to 1 gives (12). □

Proposition 15 (Phase margin)**.** *Under the same conditions,*

$$\mathrm{PM} \approx 90^\circ - \arctan\frac{f_u}{p_B} - \arctan\frac{f_u}{p_{EA}} + \arctan\frac{f_u}{z_{ESR}}. \tag{13}$$

Proof. *Step 1.* The phase of T is the sum of the phases of its factors: $-\arctan(f/p_o) - \arctan(f/p_B) - \arctan(f/p_{EA}) + \arctan(f/z_{ESR})$.

Step 2. At $f = f_u \gg p_o$, the first term is $-90^\circ + \arctan(p_o/f_u) \approx -90^\circ$.

Step 3. $\mathrm{PM} = 180^\circ + \angle T(f_u)$. □

Proposition 16 (The EA pole does not depend on the EA current)**.** *Hold V_{OV1} fixed and size M_1, M_2 by* (1)*. If C_{EA} is dominated by the drain capacitance of M_2, with c' its value per unit width, then*

$$p_{EA} \approx \frac{(\lambda_p + \lambda_n)\,K'\,V_{OV1}^2}{4\pi c' L_{12}}, \qquad A_{EA}\,p_{EA} \approx \frac{K'\,V_{OV1}}{2\pi c' L_{12}}. \tag{14}$$

Proof. *Step 1.* By Lemma 10, $p_{EA} = (\lambda_p + \lambda_n)I_{D1}/(2\pi C_{EA})$.

Step 2. By (1), $W_{12} = 2I_{D1}L_{12}/(K'V_{OV1}^2)$, so $C_{EA} \approx c'W_{12} = 2c'I_{D1}L_{12}/(K'V_{OV1}^2)$.

Step 3. Substitute Step 2 into Step 1: I_{D1} cancels, which gives the first formula.

Step 4. Multiply by $A_{EA} = (2/V_{OV1})/(\lambda_p + \lambda_n)$ (Lemmas 1 and 10). This gives the second. □

So the EA current is free; it only sets slew rate and power. Gain and bandwidth trade through V_{OV1} and L. A longer L lowers λ (more gain) but also lowers p_{EA}, which is the mechanism behind the design iterations in Part 2.

Proposition 17 (ESR window)**.** *For $f \gg z_{ESR}$, $Z_o \to R_o \| R_{ESR} \approx R_{ESR}$ and $|T| \to G g_{m,P} R_{ESR}$. The crossover stays on the single-pole slope if*

$$R_{ESR} < \frac{1}{G\, g_{m,P}}. \tag{15}$$

This condition is sufficient, not necessary. A moderate R_{ESR} adds phase lead (the $+\arctan$ term in (13)). A large R_{ESR} flattens $|T|$ above z_{ESR} and moves f_u up toward p_B and p_{EA}, so the PM collapses.

Proof. *Step 1.* By (11), $Z_o(s) \to R_o R_{ESR}/(R_o + R_{ESR})$ as $s \to \infty$, which is about R_{ESR} for $R_{ESR} \ll R_o$.

Step 2. Between z_{ESR} and $\min(p_B, p_{EA})$, $|T| \approx G g_{m,P} R_{ESR}$, which is flat.

Step 3. If that plateau is below 1, $|T|$ crossed 1 before z_{ESR}, on the $-20\,\mathrm{dB/dec}$ slope where Proposition 15 applies. □

Corollary 18 (The central trade-off)**.** *With the gate of M_P referred to ground (next section), the low-frequency PSRR is about G. By (12),*

$$f_u \approx \mathrm{PSRR}_{DC} \cdot \frac{g_{m,P}}{2\pi C_L}. \tag{16}$$

Each extra dB of PSRR raises f_u by the same factor, and f_u must stay below p_B and p_{EA}. With a single-stage EA, this is the constraint that sets the achievable PSRR.

5 Regulation, power-supply rejection and transient

Proposition 19 (Power-supply rejection)**.** *Treat V_G as referred to ground: the EA output is set by the NMOS mirror, and M_{BS} is a high impedance. Then*

$$\frac{v_{out}}{v_{in}}(s) = \frac{(g_{m,P} + g_{ds,P})\, Z_o(s)}{1 + T(s)}, \qquad \mathrm{PSRR}_{DC} \approx \frac{G}{1 + g_{ds,P}/g_{m,P}} \quad (T_0 \gg 1). \tag{17}$$

Proof. *Step 1.* The current of M_P into the output is $g_{m,P}(v_{in} - v_g) + g_{ds,P}(v_{in} - v_{out})$.

Step 2. Loop: $v_g = A_B A_{EA} \beta v_{out}$ (Proposition 9, steps 2–3 and 6).

Step 3. Output node: $v_{out}/Z_o' = g_{m,P}(v_{in} - G v_{out}) + g_{ds,P}(v_{in} - v_{out})$, where Z_o' excludes $r_{o,P}$.

Step 4. Move the $g_{ds,P} v_{out}$ term to the left. Since $1/Z_o' + g_{ds,P} = 1/Z_o$: $v_{out}(1/Z_o + g_{m,P} G) = (g_{m,P} + g_{ds,P}) v_{in}$.

Step 5. Multiply by Z_o and use $T = G g_{m,P} Z_o$. This gives the first formula.

Step 6. At DC with $T_0 = G g_{m,P} R_o \gg 1$: $v_{out}/v_{in} \approx (g_{m,P} + g_{ds,P})/(G g_{m,P})$. Invert. □

Corollary 20 (PSRR near dropout)**.** *In saturation $g_{ds,P}/g_{m,P} = \lambda V_{OV}/2$ is small and $\mathrm{PSRR}_{DC} \approx G$. Near dropout, M_P enters the triode region, and by (3) $g_{ds,P}/g_{m,P} = (V_{OV} - V_{SD})/V_{SD}$ grows. The PSRR then drops even though the EA is unchanged. The line regulation is the DC value of (17), so it degrades in the same way.*

Proposition 21 (Load-step response)**.** *On the single-pole part of the loop ($p_o \ll f \ll p_B, p_{EA}$, $R_{ESR} \to 0$), the closed-loop output impedance is $Z_{cl}(s) = Z_o/(1+T) \approx 1/\big(C_L(s + \omega_u)\big)$ with $\omega_u = 2\pi f_u$. A load step ΔI therefore gives*

$$\Delta V_{OUT}(t) = -\frac{\Delta I}{C_L\, \omega_u}\big(1 - e^{-\omega_u t}\big), \tag{18}$$

a monotone settling with time constant $\tau = 1/\omega_u = C_L/(G g_{m,P})$ and no overshoot.

Proof. *Step 1.* In this range, $Z_o \approx 1/(sC_L)$ and $T \approx \omega_u/s$ (Proposition 14).

Step 2. $Z_{cl} = \dfrac{1/(sC_L)}{1 + \omega_u/s} = \dfrac{1}{C_L(s + \omega_u)}$.

Step 3. A step $-\Delta I/s$ (current drawn from the output) gives $\Delta V(s) = -\Delta I/\big(sC_L(s + \omega_u)\big)$.

Step 4. Partial fractions: $\dfrac{1}{s(s+\omega_u)} = \dfrac{1}{\omega_u}\Big(\dfrac{1}{s} - \dfrac{1}{s+\omega_u}\Big)$. The inverse transform gives (18). □

Remark 22 (Large steps). For the $0.1 \leftrightarrow 50\,\mathrm{mA}$ step, $g_{m,P}$ itself changes by two orders of magnitude, so (18) is used piecewise. The final level comes from Corollary 6(b). The settling after a load *release* is slow, because it ends at light load where ω_u is small: $\tau_{light} = 1/(2\pi f_{u,light})$. The gate must also slew up by $\Delta V_{SG,P}$, but only with the bias current of M_{BS}, which takes $t_{slew} = C_G \Delta V_{SG,P}/I_B$. In Part 2 this is below $0.1\,\mu\mathrm{s}$, far shorter than τ_{light}.